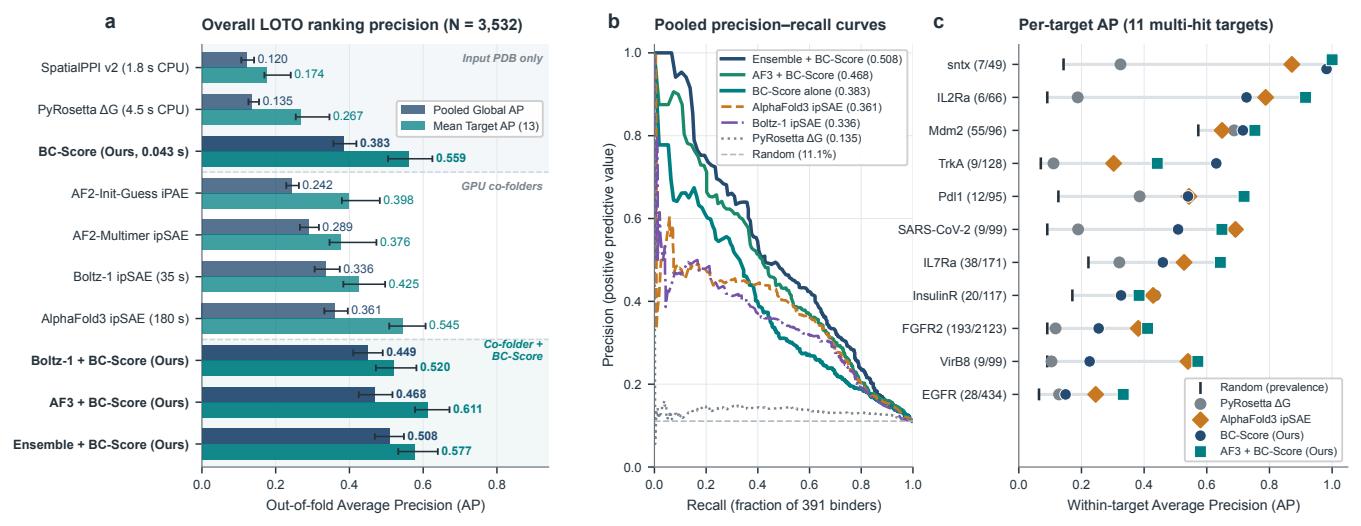

a
Overall LOTO ranking precision (N = 3,532)
Input PDB only
GPU co-folders
Co-folder + BC-Score
SpatialPPI v2 (1.8 s CPU)
PyRosetta ΔG (4.5 s CPU)
BC-Score (Ours, 0.043 s)
AF2-Init-Guess iPAE
AF2-Multimer ipSAE
Boltz-1 ipSAE (35 s)
AlphaFold3 ipSAE (180 s)
Boltz-1 + BC-Score (Ours)
AF3 + BC-Score (Ours)
Ensemble + BC-Score (Ours)
0.120
0.174
0.135
0.267
0.383
0.559
0.242
0.398
0.289
0.376
0.336
0.425
0.361
0.545
0.449
0.520
0.468
0.611
0.508
0.577
Pooled Global AP
Mean Target AP (13)
0.0
0.2
0.4
0.6
0.8
Out-of-fold Average Precision (AP)
b
Pooled precision–recall curves
Ensemble + BC-Score (0.508)
AF3 + BC-Score (0.468)
BC-Score alone (0.383)
AlphaFold3 ipSAE (0.361)
Boltz-1 ipSAE (0.336)
PyRosetta ΔG (0.135)
Random (11.1%)
Precision (positive predictive value)
Recall (fraction of 391 binders)
0.0
0.2
0.4
0.6
0.8
1.0
c
Per-target AP (11 multi-hit targets)
sntx (7/49)
IL2Ra (6/66)
Mdm2 (55/96)
TrkA (9/128)
Pdl1 (12/95)
SARS-CoV-2 (9/99)
IL7Ra (38/171)
InsulinR (20/117)
FGFR2 (193/2123)
VirB8 (9/99)
EGFR (28/434)
Random (prevalence)
PyRosetta ΔG
AlphaFold3 ipSAE
BC-Score (Ours)
AF3 + BC-Score (Ours)
0.0
0.2
0.4
0.6
0.8
1.0
Within-target Average Precision (AP)